

CMP301 Computer Networking

60% OF TOTAL COURSEWORK MARK

Release Date: 25th May 2026 - Teaching week #1

Submission Date: W/C August 3rd, 2026, Mon. at 12:00 noon – Teaching week #11

Date returned with feedback: Within 20 working days after submission: Mon., 24/08 at 12:00 noon

CW2: The SSL\TLS technology and its traffic analysis using Wireshark

CW2 is group coursework, which is worth 60% of the total coursework mark for this module.

*This coursework component requires a group of up to five students to research, write and make a presentation on the topic of the **SSL\TLS technology** and its **traffic analysis using Wireshark**. You are asked to research the topic of the assignment, prepare a PPT document and a video vodcast (e.g. a video recording) which when combined can take the form a presentation consisting of a series of PowerPoint slides; and on average each student should present his/her contribution in **approximately 7 - 8 minutes**. You should note (as you may see further below in this document) that **30% of the mark depends solely on individual contributions to the group's effort**.*

SSL stands for **Secure Sockets Layer** and, in short, it is the standard technology for keeping an internet connection secure and safeguarding any sensitive data that is being sent between two systems, preventing criminals from reading and modifying any information transferred, including potential personal details. TLS (Transport Layer Security) is just an updated, more secure, version of SSL. We still

refer to our security certificates as SSL because it is a more commonly used term. HTTPS (Hyper Text Transfer Protocol Secure) appears in the URL when a website is secured using an SSL/TLS certificate. This is carried out by making sure that any data transferred between users and sites, or between two systems remains impossible to read. That is, SSL/TLS technology uses encryption algorithms to scramble data in transit and prevent hackers from reading.

Wireshark is an open-source tool used for capturing and analysing network traffic at a very granular level. Wireshark shows packet details captured from different network sources and breaks down the traffic into the individual layers of the Open Systems Interconnection (OSI) model (e.g. the data link, network, transport, and application layers). Wireshark uses display filters to highlight and focus the analysis on the *interesting* packets .

You will be only considering the SSL\TLS protocol(s) for the analysis. This coursework component requires you to prepare a video-recorded PPT presentation (the transcript of the presentation should be a synthesis of the material you researched and may be embedded notes in a PPT documents) on SSL\TLS and its traffic analysis using Wireshark. In this coursework, it is expected that you read widely and gather your information from scientifically and technically credible sources. That is, for this CW research, you would have to read purposely and selectively.

- In general terms, for your analysis and identification of relevant concepts, approaches, theories, models, issues, challenges and understanding etc related to SSL\TLS, you may include the following points (but not limited to; few points may be overlapping): HTTPs vs HTTP, encryption and security algorithms, padlock, handshake, PKI, security certificate, history of SSL and TLS; industry standards, message authentication code, public key and private key encryption, transport layer security, authentication, hashing, deployment and Google penalised websites, Record protocol, how does SSL / TSL work?, protocol encapsulation, what is SSL / TLS, scenario of SSL work, how an SSL certificate works, digital signature, verification and trust, connection establishment, certificates vs 'good reputation'. You should note that the previous key points were not listed in a particular order; hence your presentation should be evidence based, appropriately structured and organised, and you should demonstrate critical reading and thinking skills.
- In technical terms, you should show in your analysis the initial SSL Handshake in which you should highlight the overall SSL connection behavior (e.g. you could shed a light on the several phases / steps involved in the handshake, for instance client and server send their Hellos, certificate authentication etc. If useful and adding to your explanation, you may also draw a timeline showing and naming the SSL messages sent between the client and the server during such phases / steps. You should also inspect the Hello and Certificate messages. In particular, the client list of supported ciphers and the cipher method selected at server side. You should also inspect and analyze the details of the Client Key Exchange and Change Cipher messages. This part is important as both sides will have the same secret session key (you may have to inspect the record layer to visualize the exchange of messages here). The teaching staff of the module are providing a PCAP trace file that can be used for your technical analysis of SSL\TLS using Wireshark. Your answers may be based on such a file. They are also providing a generic exercise on such a topic which would aid you in

preparing your answers and presentation; however, you do not have to show your solution of such an exercise in your presentation. Only your technical analysis is required: the exercise can provide a focus point for group work. You should prepare your PPT document and presentation in such a way that it may be understood by and useful to a fictitious group of students taking a course in computing and who may be joining placement employers.

Although you have the freedom to adopt and follow your own presentation plan and structure, the expectation, however, is that there should be an 'Introduction' in which you should cover the definition(s) and deployment(s) of SSL\TLS from which you can elaborate on the points above. Any appropriate visualization means should be used (e.g snapshots).

Your presentation should be evidence-based and supported by relevant and up-to-date references and links. Sources should include textbooks, academic web sites, manufacturers' web sites, RFCs, white papers and academic literature (conferences and journals). You may use your own selected reference style.

Note 1: You can use data from such sources as evidence, but you need to express this in your own words. Plagiarism will not be tolerated and will be dealt with according to university policy: <http://www.ulster.ac.uk/academicservices/student/plagiarism.pdf>

Note 2: You should demonstrate good knowledge and understanding of the topics and points of your presentation; and express them with high effectiveness, conciseness and succinctness. When preparing your presentation, you should make sure to include only the most relevant references.

Coursework preparation, submission and provision of feedback:

This coursework should be returned as an electronic submission by the due date specified above. University regulations require that late submissions attract a mark of zero and will be rigorously applied, without exception. If you have extenuating circumstances, you should complete an EC1 form according to your course rules; forms are found on your course website – your year tutor and course director can advise.

- What should be returned is either: The PPT document used as the basis of your presentation, the researched material can be embedded in such a document (as notes) and used as the basis that underpins your presentation (this may represent a relatively large part of your transcript; the most pertinent points should be shown in your slides). Your synthesized research material should not exceed one thousand words per group member. The presentation / vodcast (in a suitable format, e.g. mp4, you may also use Panopto etc). The individual contribution should not exceed 8 minutes in length. Those

files should be put and saved in the same folder which is then compressed (zipped); the obtained ZIP file must be uploaded using the **CW2 upload** link in the module page in the institution's VLE.

- Or as an acceptable solution is to embed your vodcast in your PPT; hence one is no longer required to add an independent vodcast. Still, your PPT slides should show the most pertinent points, and any researched material can be used as a script for the presentation; and can be added to the notes area of your PPT document.
- The length of the presentation **should not exceed 35 minutes.**
- As in the marking rubric, 75% of a student's mark depends on the collective performance of his / her group of peers. The remaining 25% of the mark depends solely on the individual contribution of the student. Hence, all group members must contribute to the presentation's collective effort. It is very important that the speaker introduces him/herself before he/she begins part of the presentation.
- **If a group member does not appear in the vodcast, then he\she should receive a null mark.**
- As it may be expected that the file to submitted can be of a substantial size, you are advised to attempt your submission early to avoid any IT related issues.